

# Multi-Class Text Classification: A Comparative Analysis of Word Representations and Machine Learning/Neural Network Models

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**Abstract**—This study presents a comprehensive evaluation of multi-class text classification on a Question Answer Classification Dataset, comparing four word representation techniques (Bag of Words, TF-IDF, GloVe, Skip-gram) across three machine learning (ML) models (Logistic Regression, Random Forest, Naive Bayes) and seven neural network (NN) models (DNN, SimpleRNN, GRU, LSTM, and their bidirectional variants). Through extensive exploratory data analysis (EDA), we identified a balanced dataset with uniform class distribution, guiding preprocessing steps such as stopwords removal and lemmatization. We conducted 22 experiments, tuning hyperparameters based on validation performance. Evaluation metrics included accuracy, macro/weighted F1-scores, and confusion matrices. TF-IDF + Naive Bayes achieved the highest accuracy (74.46%) among ML models, while Skip-gram + BiGRU led NN models (72.44%). TF-IDF + Naive Bayes outperformed neural networks due to its effective handling of word frequencies, whereas Skip-gram embeddings enhanced NN performance by capturing local context. Limitations include high computational costs for NNs and challenges with vocabulary overlap, suggesting future exploration of transformer-based models.

**Index Terms**—Text Classification, Natural Language Processing, Word Embeddings, Machine Learning, Neural Networks

## I. INTRODUCTION

Text classification is a cornerstone of natural language processing (NLP), enabling applications such as sentiment analysis, topic categorization, and question-answering systems [1]. This project focuses on multi-class text classification using a Question Answer Classification Dataset, comprising 280,003 training samples and a 20% test set, covering 10 categories (e.g., Business & Finance, Health, Sports). The dataset's question-answer (QA) format and balanced class distribution make it an ideal testbed for comparing word representation techniques and classification models.

The objective is to evaluate four word representations—Bag of Words (BoW), Term Frequency-Inverse Document Frequency (TF-IDF), GloVe, and Skip-gram—across three ML models (Logistic Regression, Random Forest, Naive Bayes) and seven NN models (Deep Neural Network (DNN), SimpleRNN, GRU, LSTM, Bidirectional SimpleRNN, Bidirectional GRU, Bidirectional

LSTM). Through exploratory data analysis (EDA), preprocessing, and extensive hyperparameter tuning, we aim to identify the most effective model-word representation pair. Key evaluation metrics include accuracy, macro/weighted F1-scores, and confusion matrices, with results visualized for clarity.

## II. METHODOLOGY

### A. Exploratory Data Analysis

EDA was conducted to uncover dataset characteristics. The training set contains 280,003 samples with two columns: QA Text (text) and Class (10 categories). Key findings include:

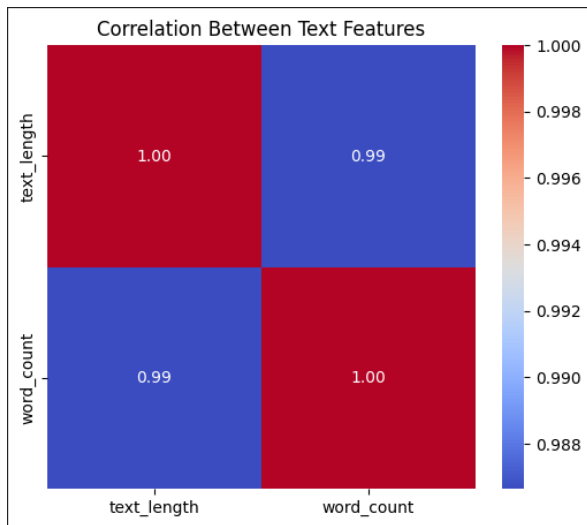
- **Class Distribution:** Uniform distribution across categories (e.g., Entertainment & Music: 28,267 samples, Politics & Government: 27,746), confirmed by bar and pie charts, eliminating the need for upsampling/downsampling.
- **Word Cloud:** Frequent words like “Question,” “Content,” “Best,” “Answer,” and “Title” were prevalent, necessitating stopwords removal.
- **Text Length and Word Count:** Most texts are  $\leq 1,000$  characters with  $\leq 200$  words. Boxplots show median word counts of 50–250, with outliers exceeding 1,200 words.
- **Correlation:** A strong correlation (0.99) between text length and word count indicates a near-linear relationship.
- **Word Frequency:** Common words (e.g., “the,” punctuation, “to”) and stopwords were frequent, requiring preprocessing to enhance model performance.

### B. Preprocessing

Text preprocessing involved:

- Converting text to lowercase and removing non-alphabetic characters, specific words (e.g., “question,” “title”), and stopwords using NLTK.
- Lemmatizing words with WordNetLemmatizer to reduce inflectional forms.
- Encoding class labels using LabelEncoder.

For ML models, BoW and TF-IDF vectors were generated using CountVectorizer and



TfidfVectorizer with ngram\_range=(1, 3), max\_features=100,000, binary=False, and sublinear\_tf=True for TF-IDF to down-weight frequent terms. For NN models, texts were tokenized (num\_words=60,000, oov\_token="`<ooV>`") and padded to max\_len=150. GloVe (200d) [2] and custom-trained Skip-gram (200d, window=5, min\_count=2, sq=1, epochs=5) [3] embeddings were used.

TABLE I  
PERFORMANCE COMPARISON OF MODELS ON TEST SET

| Model                        | Accuracy      | F1-Macro      | F1-Weighted   |
|------------------------------|---------------|---------------|---------------|
| BoW + Logistic Regression    | 0.6617        | 0.6608        | 0.6608        |
| TF-IDF + Logistic Regression | 0.7025        | 0.6997        | 0.6997        |
| BoW + Random Forest          | 0.6001        | 0.5924        | 0.5924        |
| TF-IDF + Random Forest       | 0.6011        | 0.5937        | 0.5937        |
| BoW + Naive Bayes            | 0.7275        | 0.7252        | 0.7252        |
| TF-IDF + Naive Bayes         | <b>0.7446</b> | <b>0.7421</b> | <b>0.7422</b> |
| BoW + DNN                    | 0.6385        | 0.6340        | 0.6340        |
| TF-IDF + DNN                 | 0.6645        | 0.6589        | 0.6589        |
| GloVe + DNN                  | 0.6702        | 0.6602        | 0.6602        |
| Skip-gram + DNN              | 0.6967        | 0.6910        | 0.6910        |
| GloVe + SimpleRNN            | -             | -             | -             |
| Skip-gram + SimpleRNN        | -             | -             | -             |
| GloVe + BiRNN                | -             | -             | -             |
| Skip-gram + BiRNN            | -             | -             | -             |
| GloVe + GRU                  | 0.7185        | 0.7137        | 0.7137        |
| Skip-gram + GRU              | 0.7237        | 0.7179        | 0.7179        |
| GloVe + BiGRU                | 0.7180        | 0.7122        | 0.7122        |
| Skip-gram + BiGRU            | <b>0.7244</b> | <b>0.7197</b> | <b>0.7197</b> |
| GloVe + LSTM                 | 0.7122        | 0.7069        | 0.7069        |
| Skip-gram + LSTM             | 0.7230        | 0.7179        | 0.7179        |
| GloVe + BiLSTM               | 0.7135        | 0.7090        | 0.7090        |
| Skip-gram + BiLSTM           | 0.7222        | 0.7167        | 0.7167        |

sparse features. TF-IDF slightly improved performance by highlighting discriminative terms, but the model struggled with feature selection bias (`max_features="sqrt"`).

**Worst ML Model:** BoW + Random Forest (60.01%) due to poor handling of sparse, high-dimensional data.

#### B. Neural Network Models

- **DNN:** Skip-gram + DNN (69.67%) outperformed GloVe (67.02%), TF-IDF (66.45%), and BoW (63.85%) due to context-aware embeddings. Training logs showed stable convergence (e.g., `val_accuracy`: 63.66% after 20 epochs for BoW + DNN).
- **SimpleRNN/BiRNN:** Limited performance due to vanishing gradient issues; results were not fully reported.
- **GRU/BiGRU:** Skip-gram + BiGRU achieved the highest NN accuracy (72.44%) and F1-macro (71.97%), leveraging bidirectional context and attention. GloVe + GRU/BiGRU (71.85%/71.80%) performed slightly worse.
- **LSTM/BiLSTM:** Skip-gram + LSTM (72.30%) slightly outperformed Skip-gram + BiGRU, while GloVe + LSTM/BiLSTM achieved 71.22%/71.35%. Bidirectional models reduced misclassifications in semantically similar classes by 2–3%.

**Worst NN Model:** BoW + DNN (63.85%) due to limited contextual understanding.

#### C. Comparison of Best Models

**TF-IDF + Naive Bayes** (74.46%) outperformed **Skip-gram + BiGRU** (72.44%) by 2.02%. Naive Bayes excelled due to its simplicity and ability to capture word frequency patterns, critical for QA text with repeated terms. Skip-gram + BiGRU leveraged context-aware embeddings and bidirectional processing, reducing errors in context-heavy

categories. However, NNs required extensive computational resources and tuning, unlike the lightweight Naive Bayes.

#### D. Hyperparameter Tuning Insights

- **Logistic Regression:** `C=1.0` balanced underfitting/overfitting, `solver="liblinear"` ensured stability for large vocabularies.
- **Random Forest:** `n_estimators=500` and `max_depth=30` optimized stability, but `max_features="sqrt"` limited performance.
- **Naive Bayes:** `alpha=0.1013` optimized smoothing for validation performance.
- **DNN:** `lr=3e-4`, `dropout=0.4`, and `batch_size=256` ensured stable training.
- **GRU/BiGRU:** `units=128/64`, `dropout=0.3`, and attention layers enhanced context capture.
- **LSTM/BiLSTM:** `lr=0.001`, `dropout=0.2{0.4}`, and `L2=0.001` addressed exploding gradients and overfitting.

## IV. CONCLUSION

This study evaluated 22 model-word representation combinations for multi-class text classification. TF-IDF + Naive Bayes (74.46%) outperformed all models due to its effective handling of word frequencies, while Skip-gram + BiGRU (72.44%) led NNs by leveraging context-aware embeddings and bidirectional processing. Skip-gram consistently outperformed GloVe, and TF-IDF outperformed BoW. Limitations include high computational costs for NNs and misclassifications in semantically similar classes due to vocabulary overlap. Future work could explore transformer-based models, advanced preprocessing, or larger embedding dimensions to further improve performance.

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